FATIMA ADDA-REZIG

DATA SCIENTIST · MACHINE LEARNING

Yvelines (78), Île-de-France · Available immediately · adda.fatima@hotmail.fr · GitHub · Portfolio

Biochemical engineer with **5 years of R&D experience** in recombinant protein production, purification and characterization, in an academic laboratory then in a CRO. Now a **certified Data Scientist (French RNCP Level 7, Master's level)** in machine learning, deep learning and generative AI.

> **Objective —** a Data Scientist position in biotech, pharma, CRO or medtech, where my understanding of experimental data speeds up R&D decisions.

PROFESSIONAL EXPERIENCE AND DATA SCIENCE PROJECTS

Data Scientist / Machine Learning Applied training projects · OpenClassrooms — **Mar – Dec 2026**

> **Protein solubility — ML & Deep Learning**
> DeepSol dataset audit, homology control, LightGBM/ESM-2 comparison, fine-tuning, calibration and support for experimental decisions. Gradio demo online.
> Python · PyTorch · ESM-2 · LightGBM · Optuna · MLflow · CI/CD · Gradio · Hugging Face · GitHub

> **RAG / LLM API — event search**
> Open data ingestion, FAISS indexing, semantic search, Mistral generation and source relevance control. API deployed on Render.
> Python · FastAPI · LangChain · FAISS · Mistral AI · Render · GitHub

> **Document assistant — OCR & LLM**
> OCR extraction, retrieval-augmented generation over documents and SQL data, faithfulness and answer relevance testing.
> Nanonets OCR · Mistral · SQL · FAISS · RAGAS · Streamlit · GitHub

> **Brain MRI classification — Deep Learning**
> Prototyping on partially labeled images: CNN, transfer learning and a semi-supervised approach.
> Python · PyTorch · CNN · OpenCV · GitHub

Research Associate in Biology NOVALIX · CRO, Romainville — **Oct 2024 – Jan 2026**

- **Molecular biology —** construct design, PCR, molecular cloning, DNA amplification (Maxi-Prep/Midi-Prep), transformation.
- **Cell culture —** development and adaptation of recombinant protein production protocols in various expression systems: *E. coli*, Sf9/Sf21 insect cells, HEK293-6E mammalian cells, ExpiCHO.
- **Protein biochemistry —** purification of soluble and complex membrane proteins using HPLC systems; quality control of protein samples.
- **Structural biology —** design of co-crystallization strategies and crystallogenesis according to client requests.

Research Engineer I2BC · CNRS / CEA / Université Paris-Saclay — **Dec 2017 – Feb 2021**

- Research in molecular and structural biology: protein purification, protein/DNA interactions (SEC, switchSENSE, AUC), crystallography.
- Optimization of reproducible protocols, data analysis, documentation and scientific communication.

EDUCATION

Data Scientist – Machine Learning | OpenClassrooms · RNCP Level 7 (Master's level) · 2026 · 13 projects completed

Data Scientist | DataScientest · 2021 · 400 h: Python, SQL, PySpark, ML, Deep Learning, NLP

Master's degree (M2) in Biochemistry, Molecular Biology and Biotechnology | Université d'Orléans · 2017

TECHNICAL SKILLS AND LANGUAGES

Data	Python · pandas · NumPy · SQL / PostgreSQL · exploration, analysis, data cleaning and quality control
Visualization	Matplotlib · Seaborn · Streamlit · dashboards · scientific reporting
ML & AI	scikit-learn · LightGBM · XGBoost · PyTorch · CNN · Optuna · SHAP · LLM · RAG · OCR · LangChain · FAISS · RAGAS
MLOps	MLflow · FastAPI · Docker · CI/CD · Git/GitHub · GitHub Actions · pytest · Hugging Face · Render · drift monitoring
Domain	Protein biochemistry · expression and purification · quality and traceability
Languages	French (native) · English C1 · Arabic C1